Milestone 6
Test Report and User Guide

Manual Test Cases and Usage Steps

Java Desktop Application using Swing UI and JDBC

Document Detail	Information
Course	Java Programming
Milestone	6 — Test Reports and User Guide
Marks	5
Project Title	[Enter project title]
Team Number / Name	[Enter team number or name]
Academic Year	[Enter academic year]
Submission Date	DD / MM / YYYY

Use this document as the team's submission template. Duplicate the manual test-case and task-instruction forms whenever more entries are required.

1 Submission instructions

> **Manual testing requirement**
> Students shall execute the test cases themselves and record the actual result. Automated testing, code-analysis, code-quality scoring and test-generation tools are not required for this milestone.

- Test the same project version that is submitted for evaluation.
- Record what was actually observed; do not mark a test as passed before executing it.
- Use clear test data and include negative cases, not only successful cases.
- Refer to screenshots or database observations using evidence IDs; attach the evidence in a separate annexure or folder.
- If a test fails, enter a defect and repeat the test after correction.
- Do not use real passwords, confidential information or unnecessary personal data in test evidence.

2 Team details and HackerRank progress

No.	Student name	Roll no.	Role	HackerRank profile / username	Java stars previous	Java stars current	Change
1			Team Lead				
2			Member				
3			Member				
4			Member				
5			Member				

Previous = Java stars recorded at Code Review 2. Current = Java stars on the Milestone 6 submission date. Change = Current − Previous.

3 Submission responsibility

Prepared by	[Student name(s)]
Checked by Team Lead	[Name and signature]
Project version tested	[Version, commit, build date, or file date]

PART A

Test Report

Plan, execute and record manual tests for the approved project requirements.

4 Test environment

Item	Details used for testing
Computer / operating system	
Java / JDK version	
IDE or method used to run	
Database and version	
JDBC driver / connector	
Test database name	
Application version / date	
Screen resolution, if relevant	

5 Test scope and objectives

Objective	Verify that implemented requirements work correctly through the Swing UI and that JDBC operations store, retrieve, update and delete data as expected.
Modules included	[List the modules or screens tested]
Features excluded	[List unavailable features and explain why, or write None]
Test start and end dates	From: _____ To: _____

5.1 Roles involved

Role	Name(s)	Responsibility
Test coordinator		Maintains test plan and summary
Manual tester(s)		Executes tests and records evidence
Developer(s)		Corrects defects
Retest verifier		Repeats failed tests after correction

6 Requirements-to-test traceability

Link every approved functional requirement to at least one manual test case. Use the requirement IDs from the Project Requirements Document.

Req. ID	Requirement summary	Module / screen	Test case ID(s)	Result
FR-01			TC-__	Pass / Fail / Blocked
FR-02			TC-__	Pass / Fail / Blocked
FR-03			TC-__	Pass / Fail / Blocked
FR-04			TC-__	Pass / Fail / Blocked
FR-05			TC-__	Pass / Fail / Blocked
FR-06			TC-__	Pass / Fail / Blocked
FR-07			TC-__	Pass / Fail / Blocked
FR-08			TC-__	Pass / Fail / Blocked

7 Minimum manual test coverage

Select the checks that apply to the project and create suitable test cases.

[ ] Normal workflow using valid data
[ ] Required field left empty
[ ] Invalid number, date, format or range
[ ] Duplicate value or database constraint
[ ] Create, view, update and delete operations, where applicable
[ ] Search with matching results and with no results
[ ] Cancel and confirmation actions, especially before deletion
[ ] Database unavailable or failed database operation
[ ] Close and reopen the application to verify saved data
[ ] Role or permission restrictions, if the project has user roles
[ ] Swing navigation, labels, messages and table display

8 Test data preparation

Data set	Values / records prepared	Purpose
Valid data		Normal operation
Invalid data		Validation
Boundary data		Minimum / maximum rules
Duplicate data		Constraint handling
Initial database state		Known starting point

9 Manual test-case register

Use one row for each planned test. Add rows as required, then complete a detailed test-case form for every test.

TC ID	Req. ID	Test title	Type	Tester	Date	Status	Evidence ID
TC-01	FR-__		Positive / Negative				
TC-02	FR-__		Positive / Negative				
TC-03	FR-__		Positive / Negative				
TC-04	FR-__		Positive / Negative				
TC-05	FR-__		Positive / Negative				
TC-06	FR-__		Positive / Negative				
TC-07	FR-__		Positive / Negative				
TC-08	FR-__		Positive / Negative				
TC-09	FR-__		Positive / Negative				
TC-10	FR-__		Positive / Negative				
TC-11	FR-__		Positive / Negative				
TC-12	FR-__		Positive / Negative				

10 Test execution summary

Measure	Count	Notes
Test cases planned		
Test cases executed		
Passed		
Failed		
Blocked / not executed		State the reason
Retests completed		
Pass percentage		Passed ÷ Executed × 100

Detailed Manual Test Case — TC-01

Test Case ID	TC-01	Requirement ID	FR-____
Test title	[Short action and expected purpose]	Priority	High / Medium / Low
Module / screen	[Enter module or Swing screen]	Test type	Positive / Negative / Boundary
Prepared by	[Name]	Execution date	DD / MM / YYYY

Preconditions

1.	[State the required login, screen, database state, or earlier action]
2.	

Test data

Field / input	Value used	Reason for this value

Manual execution steps

Step	Action performed by tester	Expected result after this step
1		
2		
3		
4		
5		

Final observation and evidence

Expected final result	
Actual result observed	
Evidence reference	Screenshot / database observation ID: _____
Status	[ ] Pass [ ] Fail [ ] Blocked
Defect ID, if failed	DEF-____ Retest result: [ ] Pass [ ] Fail [ ] Not retested

Detailed Manual Test Case — TC-02

Test Case ID	TC-02	Requirement ID	FR-____
Test title	[Short action and expected purpose]	Priority	High / Medium / Low
Module / screen	[Enter module or Swing screen]	Test type	Positive / Negative / Boundary
Prepared by	[Name]	Execution date	DD / MM / YYYY

Preconditions

1.	[State the required login, screen, database state, or earlier action]
2.	

Test data

Field / input	Value used	Reason for this value

Manual execution steps

Step	Action performed by tester	Expected result after this step
1		
2		
3		
4		
5		

Final observation and evidence

Expected final result	
Actual result observed	
Evidence reference	Screenshot / database observation ID: _____
Status	[ ] Pass [ ] Fail [ ] Blocked
Defect ID, if failed	DEF-____ Retest result: [ ] Pass [ ] Fail [ ] Not retested

11 Defect and retest log

Create a defect for every failed test. Severity describes the effect on the user; it is not a judgment about the person who wrote the code.

Defect ID	TC ID	Problem observed	Severity	Assigned to	Status	Retest
DEF-01	TC-__		Critical / Major / Minor		Open / Fixed	Pass / Fail
DEF-02	TC-__		Critical / Major / Minor		Open / Fixed	Pass / Fail
DEF-03	TC-__		Critical / Major / Minor		Open / Fixed	Pass / Fail
DEF-04	TC-__		Critical / Major / Minor		Open / Fixed	Pass / Fail
DEF-05	TC-__		Critical / Major / Minor		Open / Fixed	Pass / Fail
DEF-06	TC-__		Critical / Major / Minor		Open / Fixed	Pass / Fail

11.1 Severity guide

Severity	Meaning
Critical	Application cannot start, data is seriously corrupted, or a main workflow cannot continue.
Major	An important feature gives a wrong result, but other parts of the application can still be used.
Minor	A small usability, display, wording or low-impact functional issue.

12 Test conclusion

Overall result	[ ] Ready [ ] Ready with limitations [ ] Not ready
Important verified workflows	
Open defects / limitations	
Database consistency checked	[ ] Yes [ ] No Method used: _____
Team Lead conclusion	

PART B

User Guide

Explain how a first-time user can set up, start and use the application safely.

13 Application overview

Application name	
Purpose	[Explain the problem solved in two or three sentences]
Intended users	[List user types or roles]
Main functions	[List the major tasks supported]
Known limitations	[State current limitations honestly, or write None]

14 Requirements before installation

Requirement	Required version / detail	Where to obtain or configure
Operating system		
Java / JDK		
Database server		
JDBC driver		
IDE or runnable JAR		
Database script / sample data		
Other library, if any		

15 Files supplied

File / folder	Purpose

16 Installation and database setup

Replace the guidance below with the exact steps for your project. A user should not need to inspect the source code to discover the setup process.

Step	Instruction	Expected result / check
1	Copy or extract the submitted project to a suitable folder.	Project files are visible.
2	Install or start the required database server.	Database service is running.
3	Create the application database and run the supplied schema or import script.	Required tables are created.
4	Configure the database URL, username and password in the stated configuration location.	Configuration points to the test database.
5	Add or verify the required JDBC driver and other project libraries.	Project builds without missing-library errors.
6	Open the project in the stated IDE, or locate the supplied runnable JAR.	Project or JAR is ready to start.
7	Run the main class or JAR and wait for the first Swing screen.	Application opens successfully.

17 Starting and closing the application

Action	Steps	Expected result
Start		
Log in, if applicable		
Log out, if applicable		
Close safely		

18 Screen and navigation guide

Give each screenshot an evidence reference such as UG-01. Use cropped screenshots that show the relevant controls clearly.

Screen / module	Purpose	How to open	Important controls	Screenshot ID
				UG-01
				UG-02
				UG-03
				UG-04
				UG-05
				UG-06

18.1 Navigation map

Main screen → _____ → _____ → _____

Explain how the user returns to the previous screen or main menu:

_____

19 User input conventions

Input / symbol	Required format or meaning	Example
Required fields		
Date		
Number / amount		
Identifier / code		
Search term		
Confirmation message		

20 Major user tasks

Write one task sheet for every major activity, for example: add a record, search or view, update, delete with confirmation, perform a transaction, and generate a report. Duplicate the following form as needed.

Task Instruction Sheet — TASK-01

Task name	[Enter a user goal, such as Add a new student]
User / role	
Starting screen	
Preconditions	

Usage steps

Step	What the user should do	What the user should see
1		
2		
3		
4		
5		
6		
7		

Completion and exceptions

Successful result	
Validation / error messages	
How to cancel or undo	
Screenshot reference(s)	UG-____

Writing rule
Begin each instruction with a clear action such as Select, Enter, Click, Choose or Confirm. State the exact button and field labels used in the Swing interface.

Task Instruction Sheet — TASK-02

Task name	
User / role	
Starting screen	
Preconditions	

Usage steps

Step	What the user should do	What the user should see
1		
2		
3		
4		
5		
6		
7		

Completion and exceptions

Successful result	
Validation / error messages	
How to cancel or undo	
Screenshot reference(s)	UG-____

21 Messages and troubleshooting

Message / symptom	Likely cause	Action for the user
Application does not start		
Database connection fails		
Required-field message appears		
Invalid number or date		
Duplicate record message		
No search result		
Save / update / delete fails		
Other: _____		

22 Data safety, backup and recovery

- Location of the application database or backup file: _____
- Steps to create a backup: _____
- Steps to restore a backup: _____
- Data that should not be deleted without confirmation: _____
- Contact or action when recovery is not possible: _____

23 User-guide verification checklist

[ ] A first-time user can install or run the application using only this guide.
[ ] Every major user task has numbered steps and an expected result.
[ ] Screen names, button labels and screenshots match the submitted application.
[ ] Input formats and required fields are explained.
[ ] Common errors and corrective actions are included.
[ ] Setup steps include Java, database and JDBC requirements.
[ ] No real password or confidential test data appears in the document.

24 Final submission checklist and confirmation

Item	Included	File / evidence reference
Completed Test Report	[ ]	
Manual test-case evidence / screenshots	[ ]	
Defect and retest record	[ ]	
Completed User Guide	[ ]	
Database schema / setup script	[ ]	
Sample data, if required	[ ]	
Source code / runnable version tested	[ ]	
HackerRank profiles and current Java stars	[ ]	Section 2

24.1 Team declaration

We confirm that the manual test results recorded in this document are based on tests executed on the submitted project version. The user guide describes the same application version and its actual usage steps.

Role	Name	Signature	Date
Team Lead			
Team Member			
Team Member			
Team Member			
Team Member			

24.2 Faculty verification

Comments	
Marks awarded	_____
Faculty name and signature	
Date	

End of template